SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO1: *Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 20 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I SHAGITHIYAN N 192324259 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1.

1. **Identifying students who may fail:**
   Data Science can analyze students' attendance, marks, assignments, and past performance to **predict students who are at risk of failing**.
2. **Hospital patient records:**
   Data Science can analyze historical patient data to **predict diseases, improve diagnosis, and support better treatment decisions**.
3. **Online shopping recommendations:**
   This is an example of a **Recommendation System**, which suggests products based on customers' previous purchases and interests.
4. **Detecting suspicious bank transactions:**
   Data Science can be used for **fraud detection** by identifying unusual transaction patterns.
5. **Reducing customer loss:**
   Data Science can perform **customer churn prediction** to identify customers who may leave and help the company take preventive action.
6. **Improving student performance:**
   One benefit is **personalized learning**, where student data can be used to identify learning needs and provide suitable support.
7. **Festival product sales:**

Data Science can analyze sales data to identify **seasonal and festival buying patterns**, helping the supermarket plan inventory.

8. **Reducing traffic congestion:**
   Data Science can analyze traffic, GPS, and road data to **predict congestion and optimize traffic routes/signals**.
9. **Improving crop yield:**
   Data Science can analyze weather, soil, and crop data to **predict the best time for planting and improve crop yield**.
10. **Excel tables, reviews, images, and videos:**
    This shows **variety**, one of the characteristics (facets) of Big Data.
11. **Temperature updated every minute:**
    This represents **velocity**, because data is being generated and updated rapidly.
12. **University data in rows and columns:**
    This is **structured data**.
13. **Customer comments on social media:**
    These are generally **unstructured data** because they can contain free-form text, images, videos, etc.
14. **Very large amount of daily app activity data:**
    This represents **volume**, the large quantity of data being generated.
15. **Missing ages and incorrect phone numbers:**

This indicates a **data quality issue**, specifically **incomplete and inaccurate data**.

16. **Converting customer data into useful sales decisions:**
The important Big Data characteristic is **value**—extracting useful information and business value from data.

17. **Storing large files using Hadoop Distributed File System:**
The component used is **HDFS (Hadoop Distributed File System)**, which provides distributed storage.

18. **Processing large datasets using parallel computation:**
Apache Spark can be used for fast, distributed, parallel processing of large datasets.

19. **Collecting live IoT sensor data:**
Apache Kafka is a suitable Big Data ecosystem component for collecting and streaming real-time data.

20. **Visualizing Big Data results:**
Data visualization tools such as Tableau or Power BI can be used to present Big Data results through charts, dashboards, and graphs.